Octavio Morales
CS503 – Homework 1

1.1. a. $P(J|S) = \frac{P(J \cap S)}{P(S)} = \frac{0.08}{0.30} \approx 0.267$ $P(J|S) \approx 0.267$

b. $P(J|\overline{S}) = \frac{P(J) - P(J \cap S)}{1 - P(S)} = \frac{0.2 - 0.08}{1 - 0.3} \approx 0.171$ $P(J|\overline{S}) \approx 0.171$

c. $P(J \cap S | J \cup S) = \frac{P(J \cap S)}{P(J) + P(S) - P(J \cap S)} = \frac{0.08}{0.2 + 0.3 - 0.08} \approx 0.190$

$P(J \cap S | J \cup S) \approx 0.190$

1.2. H = Harold S = Sharon

a. $P(H \cap \overline{S}) = P(H) - P(H \cap S) = 0.80 - 0.79 = 0.01$

$P(H \cup S) = P(H) + P(S) - P(H \cap S)$ $P(H \cap \overline{S}) = 0.01$

$0.91 = 0.8 + 0.9 - P(H \cap S)$

$P(H \cap S) = 0.79$

1.2. b. $P(S\cap\overline{H})=P(S)-P(H\cap S)=0.9-0.79=0.11$

$P(S\cap\overline{H})=0.11$

c. $P(\overline{H}\cap\overline{S})=1-P(H\cup S)=1-0.91=0.09$

$P(\overline{H}\cap\overline{S})=0.09$

1.3. J = Jerry is at the bank
S = Susan is at the bank

$P(J\cap S)=P(J)\cdot P(S)=0.2\cdot 0.3=0.06$

The events are not independent because the calculated $P(J\cap S)$ is not equal to the given $P(J\cap S)$.

1.4. a. Sum of 6 Possibilities: (1,5), (2,4), (3,3), (4,2), (5,1) => 5/36
D_1 = The sum is 6
D_2 = The second die shows 6

$P(D_1\cap D_2)=1/36$ <= only (1,5) works $1/36 \neq 5/216$
$P(D_1\cap D_2)=P(D_1)\cdot P(D_2)=5/36\cdot 1/36=5/216$

The two events (D_1 and D_2) are not independent.

1.4 b. D_1 = the sum is 7
D_2 = the first dice is 5

Sum of 7 Possibilities: $(1,6),(2,5),(3,4),(4,3),(5,2),(6,1) \Rightarrow 6/36 = 1/6$

$P(D_1 \cap D_2) = 1/36 \Leftarrow$ only $(5,2)$ works $\quad 1/36 = 1/36$
$P(D_1 \cap D_2) = 1/6 \cdot 1/6 = 1/36$

The two events (D_1 and D_2) are independent.

1.5. TX = Texas, AK = Alaska, NJ = New Jersey, I = Oil

1. $P(I) = P(I|TX)\cdot P(TX) + P(I|AK)\cdot P(AK) + P(I|NJ)\cdot P(NJ)$
$P(I) = 0.3\cdot 0.6 + 0.2\cdot 0.3 + 0.1\cdot 0.1$
$P(I) = 0.25$

2. $P(TX|I) = \dfrac{P(I|TX)\cdot P(TX)}{P(I)} = \dfrac{0.3\cdot 0.6}{0.25} = 0.72$

$P(TX|I) = 0.72$

1.6.1. $P(\text{Not Survived}) = \frac{\Sigma(\text{Not Survived})}{\text{Total}} = \frac{1490}{2201} \approx 0.677$

$P(\text{Not Survived}) \approx 0.677$

2. $P(\text{1st Class}) = \frac{\Sigma(\text{1st Class})}{\text{Total}} = \frac{325}{2201} \approx 0.148$

$P(\text{1st Class}) \approx 0.148$

3. $P(\text{1st Class} \mid \text{Survived}) = \frac{\text{1st Class} \cap \text{Survived}}{\Sigma(\text{Survived})} = \frac{203}{711} \approx 0.285$

$P(\text{1st Class} \mid \text{Survived}) \approx 0.285$

4. $P(\text{Survived}) = \frac{711}{2201} \approx 0.323$

$P(\text{1st Class}) \cdot P(\text{Survived}) = 0.148 \cdot 0.323 \approx 0.048$

$P(\text{1st Class} \cap \text{Survived}) = \frac{203}{2201} \approx 0.092 \quad 0.092 \neq 0.048$

The events are not independent.

5. $P(\text{1st Class and Child} \mid \text{Survived}) = \frac{\text{1st Class Child} \cap \text{Survived}}{\Sigma(\text{Survived})}$

$P(\text{1st Class Child} \mid \text{Survived}) = 0.008 \quad = \frac{6}{711} = 0.008$

1.6. 6. $P(\text{Adult} | \text{Survived}) = \frac{\text{Adults} \cap \text{Survived}}{\Sigma(\text{Survived})} = \frac{654}{711} \approx 0.920$

$P(\text{Adults} | \text{Survived}) \approx 0.920$

7. $P(\text{Adult} \cap \text{1st Class} | \text{Survived}) = \frac{197}{711} \approx 0.277$

$P(\text{Adult} | \text{Survived}) \cdot P(\text{1st Class} | \text{Survived}) = 0.92 \cdot 0.285 \approx 0.262$

$0.277 \neq 0.262$

The two events are not independent.

1.7. TP = True Positive, TN = True Negative, FP = False Positives, FN = False Negatives

Accuracy $= \frac{TP + TN}{\text{Total}} = \frac{970 + 930}{2000} = \frac{1900}{2000} = 0.95$ Accuracy $= 0.95$

Precision $= \frac{TP}{TP + FP} = \frac{970}{970 + 70} \approx 0.933$ Precision $= 0.933$

Recall $= \frac{TP}{TP + FN} = \frac{970}{970 + 30} = 0.97$ Recall $= 0.97$

F1 $= 2 \cdot \left(\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}\right) = 2 \cdot \frac{0.933 \cdot 0.97}{0.933 + 0.97} \approx 0.951$ F1 $= 0.951$